

A defensible delivery path

Immutable deployment, private telemetry, and evidence-led operations

DEPLOYMENT

Immutable

GHCR digest deployed through strict SSH with candidate validation and rollback retention.

MONITORING

Private

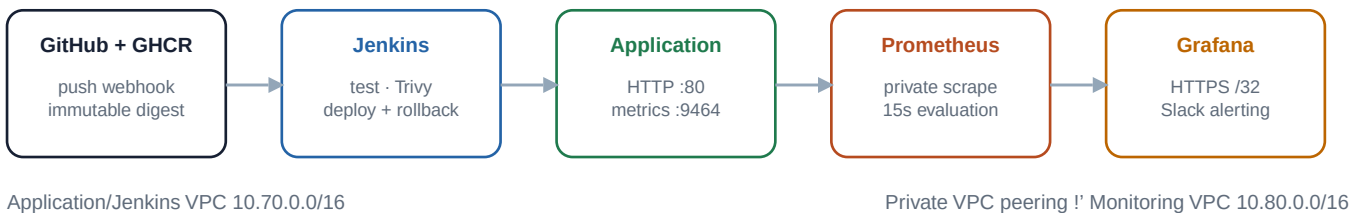
Prometheus and Node Exporter stay off the public network; Grafana HTTPS is admin-restricted.

VERIFICATION

Controlled

>5% 5xx test proved Pending !' Firing !'
Resolved, then restored normal service.

ARCHITECTURE AT A GLANCE



CONTROLS THAT MATTER

Delivery safeguards

Jenkins runs syntax/Jest/JUnit and repository/image Trivy gates before push. The application runs as UID/GID 10001 with a read-only filesystem, dropped capabilities, no-new-privileges, resource limits, and no Docker socket.

Telemetry and secrets

Only Grafana is exposed. Application metrics and both Node Exporters use private peering. Slack and Cloudflare credentials are fetched at runtime from scoped Secrets Manager entries—not from Git, Terraform state, logs, or evidence.

SIGNALS AND ALERT POLICY

The provisioned dashboard tracks request rate, p95 latency, HTTP 5xx percentage, CPU, memory, disk, and scrape health.

Prometheus evaluates three recording rules every 15 seconds. Grafana is the sole alerting engine: JenkinsWebappHighErrorRate requires both a 5xx rate above 5% for five minutes and at least 20 requests in that window, then sends firing and resolved notifications to Slack.

No raw URLs, request bodies, identities, tokens, or query strings are used as metric labels.

Recovery was observed end to end

The test validates alerting and logging without presenting controlled traffic as an incident

CONTROLLED ALERT LIFECYCLE · 2026-08-30 UTC

●	14:04:28	Traffic started	Normal and deliberate fault requests began every five seconds.
●	14:04:55	Pending	JenkinsWebappHighErrorRate entered Pending.
●	14:10:48	Firing	The threshold remained true for the required five minutes.
●	14:11:00	Traffic stopped	No further controlled requests were generated.
●	14:12:14	Fault disabled	Controller disabled; fault route returned HTTP 404.
●	14:15:52	Resolved	The high-error alert returned to its normal state.

WHAT THE VERIFICATION PROVES

RECOVERY STATE

4 / 4 UP

Application, application-node, monitoring-node, and Prometheus were UP after restoration. The normal container was healthy, non-root, read-only, and fault injection was absent.

LOG CORRELATION

156 events

CloudWatch Logs Insights matched 156 controlled application 500/error events from 14:04:28.862 to 14:10:55.089 UTC, bracketing the traffic window.

AUDIT POSTURE, LIMITATIONS, AND CLEANUP

CloudTrail and GuardDuty

The dedicated archive has multi-Region management events, log-file validation, KMS encryption, blocked public access, versioning, and 30/90/365-day lifecycle controls. GuardDuty is enabled; the saved finding is AWS-generated synthetic evidence.

Limits and teardown boundary

Monitoring runs on a cost-sensitive t3.micro. Official stable Grafana, Node Exporter, and Nginx images retain a documented lab exception for fixable Trivy HIGH findings; no CRITICAL finding is accepted. Preserve evidence, refresh ownership, then apply only an explicitly approved destroy plan. Shared resources are excluded.

EVIDENCE INDEX

Phase 11 live verification · Phase 6 CloudTrail verification · Phase 7 GuardDuty synthetic sample · evidence/README.md